

Teacher:	Ray Joshua Balingkit	**Subject:**	Computing
Week:	3 (Term 1)	**Year:**	7
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will learn to add unsigned binary numbers using carry bits and check a result in decimal. They will also learn to convert decimal values to eight-bit unsigned binary and recognise when a result exceeds eight bits. Through teacher demonstrations, discussion and practical activities, students will practise these skills and explain their ideas.

End-of-Week Assessment/Check: How will you know they got it?

Use teacher observation, pupil explanations and the completed practical or paper activity. Check the right-to-left addition, carry placement and a decimal verification rather than accepting an unexplained answer. Check an independent conversion and an explanation that discarding the ninth bit loses the correct unsigned result. Note pupils who need further guided practice.

LESSON SLOTS - MONDAY AND TUESDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	7 September 2026	By the end of this lesson, students will be able to add unsigned binary numbers using carry bits and check a result in decimal.	**H:** Ask what happens when 1 and 1 are added in base 2. **ID:** Introduce 0+0, 0+1, 1+1 and 1+1+1, linking each result to a carry into the next column. **WD:** Work through 0011 + 1011 = 1110, showing each carry and checking 3 + 11 = 14. **YD:** Pupils complete four-bit additions, show carries and verify one result by converting to decimal; extend to eight bits when secure. **C:** Check the right-to-left addition, carry placement and a decimal verification rather than accepting an unexplained answer.	**Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing. **Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
2	8 September 2026	By the end of this lesson, students will be able to convert decimal values to eight-bit unsigned binary and recognise when a result exceeds eight bits.	**H:** Display eight place-value cards and ask which combine to make 20. **ID:** Explain selecting powers of two, the	**Support:** Provide place-value grids, carry rows, a character-code table and labelled diagrams. Model one step at a time and allow spoken reasoning before writing.

			unsigned eight-bit range 0 to 255 and overflow beyond that range. **WD:** Convert 20 to 00010100 and check it back; add 11111111 and 00000001 to show a ninth bit is needed. **YD:** Pupils convert selected values from 0 to 255, check their sums and explain an example that cannot fit in eight bits. **C:** Check an independent conversion and an explanation that discarding the ninth bit loses the correct unsigned result.	**Extension:** Ask pupils to justify a method, identify a deliberate error or test an eight-bit boundary, then verify the result using another representation.
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Oxford International Lower Secondary Computing Student Book 7 printed pages 14-21
- Teacher computer and display
- Classroom computers or laptops and the class save folder
- Binary place-value grids, carry rows, character-code tables and pixel or sampling sheets
- Whiteboard and markers
- Student notebooks, pencils and task sheets
- Prepared examples and the lesson success checklist



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?